DAY-2

TCP/IP Model and Packet Analysis Assignment

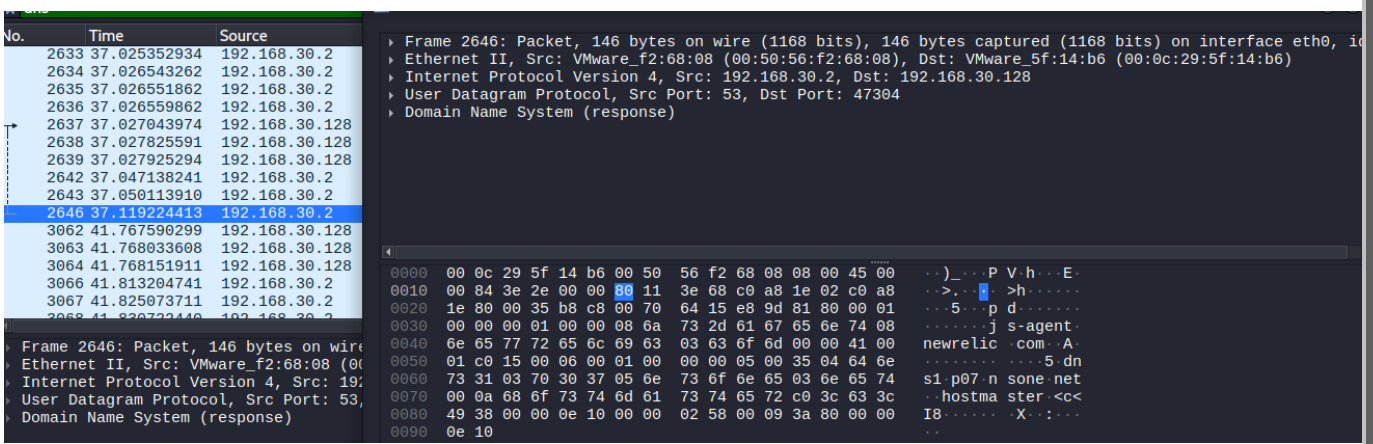
Q1. Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

Answer:

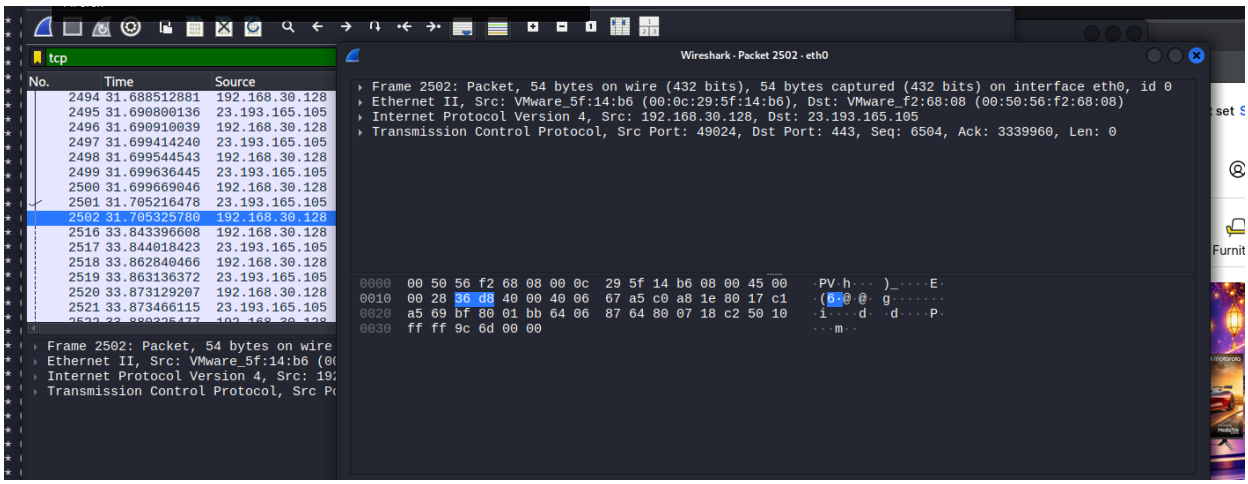
1. **Start capture:** Open Wireshark and double-click the interface eth0 (the one with the moving graph).
2. **Refresh Flipkart:** In Firefox open flipkart.com and press the refresh button (F5).
3. **Stop capture:** Click the red Stop button.
4. **Find the packets:** Use the display filter box (type the filter and press Enter) and click a packet. In the middle pane, open its headers to see each layer: Ethernet II, Internet Protocol, Transmission Control Protocol and the application data.

Filters to use: dns (Application), tcp (Transport), ip (Internet), arp or eth (Network Access).

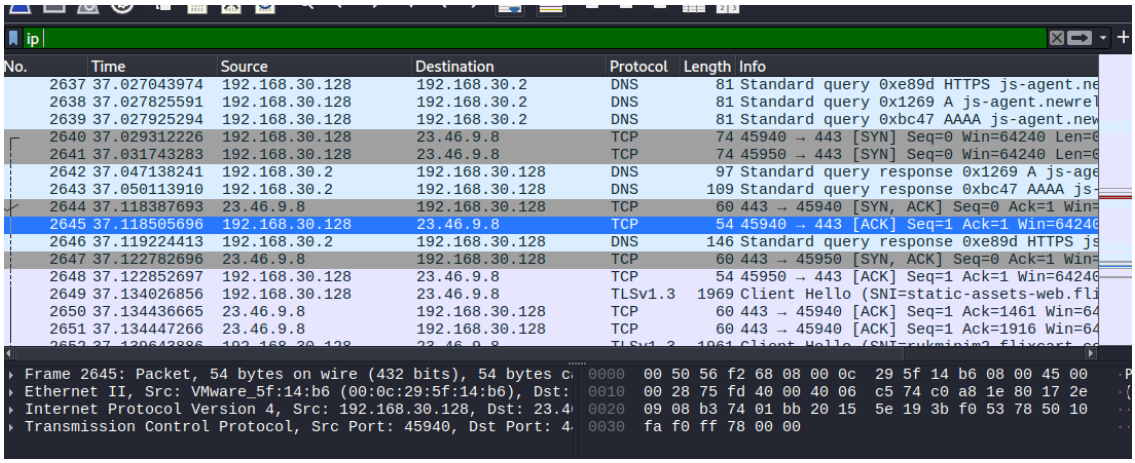
TCP/IP layer	Packet no.	Protocol	How I identified it (header)
Application	2646	DNS	The Domain Name System (response) header is at the top of the packet, carried over UDP (Source Port 53, Destination Port 47304). Info: Standard query response from the DNS server 192.168.30.2.
Transport	2502	TCP	The Transmission Control Protocol header shows Source Port 49024 and Destination Port 443 (HTTPS), with Sequence number 6504 and Acknowledgment number 3339960. Len: 0 means it carries no data, only a confirmation.
Internet	2645	IPv4	The Internet Protocol Version 4 header shows Source IP 192.168.30.128 (my Kali machine) and Destination IP 23.46.9.8 (server). Info: 45940 > 443 [ACK].
Network Access	2503	ARP (Ethernet II)	The Ethernet II header shows Source MAC VMware_c0:00:08 and Destination Broadcast (ff:ff:ff:ff:ff:ff). The ARP request asks for the MAC address of a device on the local network.



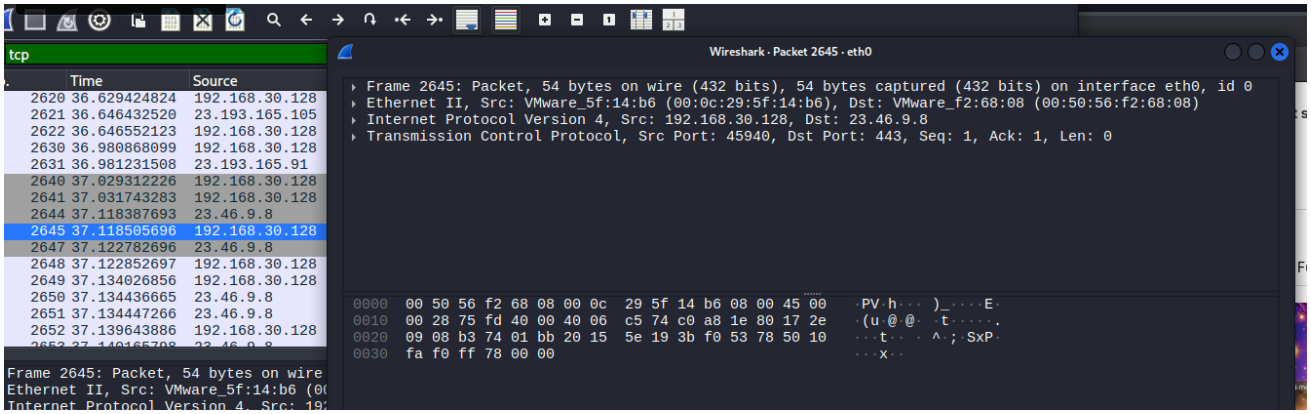
Packet 2646: DNS (Application layer), header window



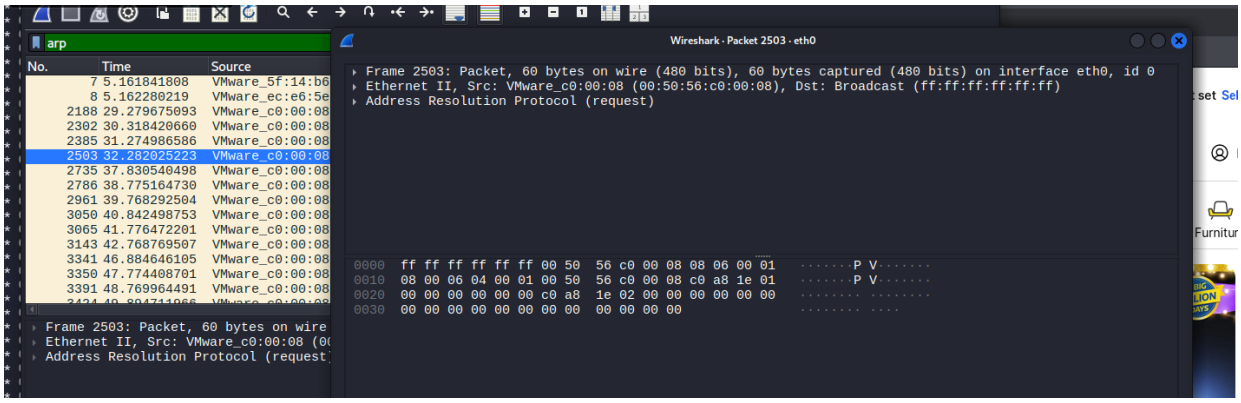
Packet 2502: TCP (Transport layer), header window



Packet 2645: IPv4 (Internet layer), filter: ip



Packet 2645: header window showing Source IP 192.168.30.128 and Destination IP 23.46.9.8



Packet 2503: ARP / Ethernet II (Network Access layer), header window

Q2. Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

Answer:

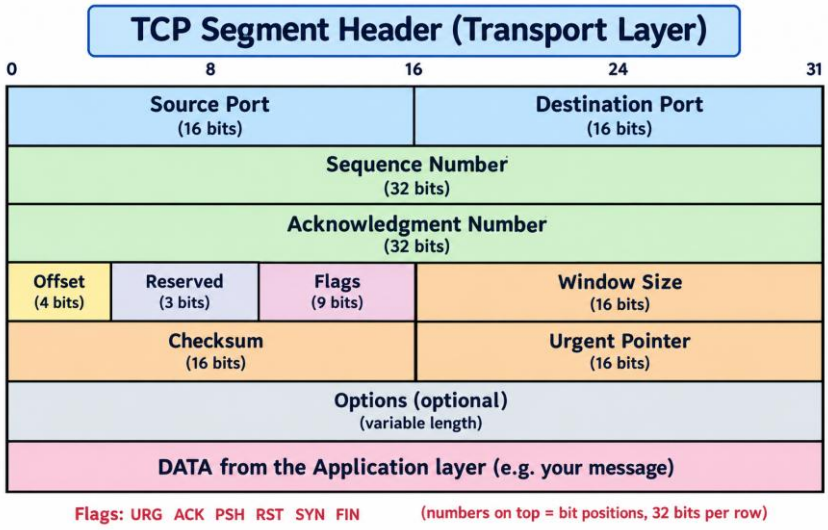
OSI no.	OSI layer	TCP/IP layer	Real-world example (app)	Example detail
7	Application	Application	WhatsApp message	The app creates and sends the message
6	Presentation		Instagram photo	Photo is compressed (JPEG) and encrypted
5	Session		Flipkart login	You stay logged in while shopping
4	Transport	Transport	Zomato order	TCP makes sure the full order reaches the server
3	Network	Internet	Opening Instagram	IP address and routers find the server
2	Data Link	Network Access	Phone connecting to Wi-Fi	Wi-Fi frames with MAC address
1	Physical		Mobile data / Wi-Fi signal	Radio waves and cables carry the bits

The 7 OSI layers are grouped into 4 TCP/IP layers. Application, Presentation and Session together form the TCP/IP Application layer, and Data Link and Physical together form the Network Access layer.

Q3. Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

Answer:

The TCP segment header is shown below with all its fields labeled. Every field of the TCP header belongs to the Transport layer. The data after the header comes from the Application layer.



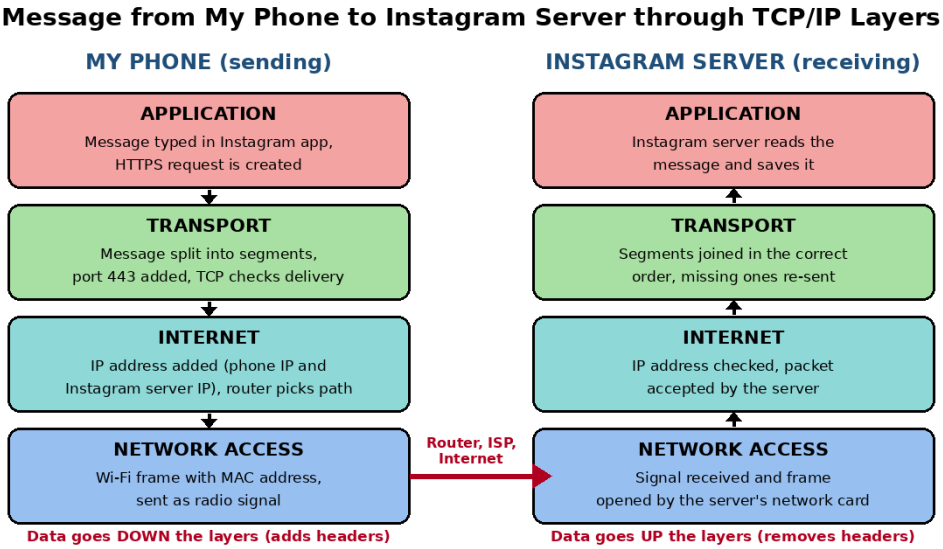
Field	Size	What it does
Source Port	16 bits	The port number of the sending app/device (e.g. 32986). It tells which program sent the data.
Destination Port	16 bits	The port number of the receiving service (e.g. 443 for HTTPS). It tells which program on the server gets the data.
Sequence Number	32 bits	The number of the first byte in this segment. It keeps the pieces of data in the correct order.
Acknowledgment Number	32 bits	The next byte the receiver expects. It confirms that earlier data arrived safely.
Data Offset (Header Length)	4 bits	Shows how long the TCP header is, so the receiver knows where the data starts.
Reserved	6 bits	Kept for future use and normally set to 0.
Flags (URG, ACK, PSH, RST, SYN, FIN)	6 bits	Control the connection: SYN starts it, ACK confirms data, FIN ends it, RST resets it, PSH sends data quickly, URG marks urgent data.
Window Size	16 bits	How much data the receiver can accept at once. It controls the speed (flow control).
Checksum	16 bits	Used to check if the header and data got damaged on the way (error checking).
Urgent Pointer	16 bits	Points to urgent data when the URG flag is set. It is rarely used.
Options	Variable	Extra settings, such as the maximum segment size, used while making the connection.

Values from my own Wireshark capture (packet 2502):

Field	Value in packet 2502	Meaning
Source Port	49024	The port my machine used to send the data
Destination Port	443	The web server's HTTPS port
Sequence Number	6504	Position of the data my machine has sent
Acknowledgment Number	3339960	Next byte my machine expects from the server
Length (Len)	0	No data inside, only a confirmation

Q4. Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).

Answer:



Q5. Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

Answer:

Packet picked: Packet no. 2090 from my capture, Info: 32986 > 443 [ACK] Seq=2137 Ack=5257 (source 192.168.30.128, destination 34.107.243.93).

- **TCP/IP layer:** Transport layer.
- **How I identified it:** The Protocol column shows TCP. In the middle pane the Transmission Control Protocol header shows the Source Port 32986, Destination Port 443 and the ACK flag with sequence and acknowledgment numbers. These fields belong to the Transport layer.
- **Real-world action:** Port 443 means HTTPS, so this was a website or app connection. The ACK packet is my device telling the server "I received your data". It happens when a page or feed loads, for example when Instagram sends photos and my phone confirms that each part arrived.

